EGPO: Enhanced Guidance and Pseudo-Label Optimization for Semi-Supervised Semantic Segmentation of Remote Sensing Images

Xifeng Xue, Hao Zhu, *Member, IEEE*, Xiaotong Li, Jianda Wang, Longsheng Qu, and Biao Hou, *Senior Member, IEEE*

***Abstract*—Owing to complex remote sensing image features boosting manually labeled costs, semi-supervised semantic segmentation learns from limited labeled and abundant unlabeled data to alleviate this dilemma. However, there are still many challenges in the practical application of this technology, such as incorrect unsupervised information misleading the model and causing errors in pseudo-labels, thereby causing error accumulation. This article proposes a semi-supervised semantic segmentation method for remote sensing images. The enhanced guided learning module we designed uses a label-guided model to predict unlabeled data in the correct direction, enriching ground features and unsupervised information and alleviating the problems of misprediction and consistency regularization failure caused by the lack of labels. At the same time, facing the noise generated during the data augmentation process, our designed unsupervised loss dynamic screening module aims to locate and suppress the noise adaptively. In addition, in the face of inevitable erroneous predictions in pseudo-labels, we design a pixel category selection (PCS) module that produces high-quality and high-confidence pseudo-labels through multistep filtering and dual model fusion. Ultimately, by conducting experiments on the DFC22, iSAID, MER, GID-15, and Vaihingen datasets, we successfully verified the effectiveness of our proposed method. The source code has been made public: https://github.com/Xidian-AIGroup190726/EGPO**

***Index Terms*—Consistency regularization, remote sensing, self-training, semi-supervised semantic segmentation.**

I. Introduction

THE various remote sensing images with different resolutions can now be readily acquired and applied with the rapid development of satellite and remote sensing technologies [1], [2], [3]. Semantic segmentation of remote sensing images has begun to be applied in various fields [4], [5], [6]. Semantic segmentation is a crucial and indispensable computer vision task that accurately classifies images. It assists in land classification, road extraction, vegetation cover detection, land degradation assessment, and earthquake, fire, and flood detection. It has an exceptionally significant assisting function in fields such as national security, urban planning, and other aspects [7], [8], so we need to fully and reasonably utilize remote sensing images [9], [10].

Although supervised learning-based semantic segmentation of remote sensing images has achieved considerable success [11], [12], supervised learning means that each high-precision image requires accurate labels manually labeled, and pixel-level labeling is time-consuming and expensive [13], which also limits the wide application of semantic segmentation in remote sensing images. To overcome this limitation, semi-supervised learning is gradually being applied to semantic segmentation, allowing models to learn with limited labeled data and large amounts of unlabeled data. The three training methods currently available are GAN, self-training, and consistency regularization.

In the early stages of semi-supervised semantic segmentation development, most research focused on generative networks, represented by the GAN framework [14], [15], [16], which used discriminative networks to obtain additional information from unlabeled data. On the basis of this, Mittal et al. [17] designed two branch networks to replace the traditional GAN loss, making the predicted segmentation map by the generator consistent with the real segmentation map in terms of feature statistics and stabilizing adversarial training. Due to unstable training, high data sensitivity, and huge computational costs, semi-supervised semantic segmentation relying solely on generative networks has gradually declined. The current mainstream approach is to deeply couple it with consistency regularization and other strategies. The LeakGAN [15] combined with consistency regularization, a semi-supervised retinal vessel segmentation framework that combines information leakage contamination discriminator with mean teacher regularization, achieving high-precision segmentation with minimal annotation.

Self-training [18], [19], [20], [21], [22], [23], [24] methods have been applied to semi-supervised semantic segmentation, which is the process of generating pseudo-labels by predicting unlabeled data and combining them with labeled data to train a model. Since the introduction of semi-supervised semantic segmentation using pseudo-labeling [25], the confidence level of pseudo-labels is closely related to the final segmentation accuracy. Therefore, how to obtain high confidence pseudo-labels has become a key bottleneck for breakthroughs in this

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an 710126, China (e-mail: haozhu@xidian.edu.cn).

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field. ST++ [26] adopts an improved self-training method by constructing a three-stage self-training framework to separate unlabeled data into simple and difficult samples for processing, further improving the confidence of pseudo-labels. Considering the complex terrain and land features of remote sensing images, it is difficult to generate prediction errors when generating pseudo-labels. If pseudo-labels are learned without filtering, it inevitably leads to the model training in the wrong direction. For remote sensing images, the recently proposed LSST [27] uses adaptive methods to obtain reliable regions through classification confidence thresholds, greatly improving the confidence of pseudo-labels. However, labeling is extremely scarce, and once the model encounters unlabeled data outside the distribution, it is bound to generate a large number of misclassifications. Threshold filtering alone is difficult to eliminate the noise in the pseudo-labels, thus seriously weakening their credibility.

On the other hand, consistency regularization techniques have also been applied to semi-supervised semantic segmentation tasks [28], [29], [30], [31], [32], [33], [34], [35]. The consistency regularization method [32] can utilize unlabeled data without generating pseudo-labels. By forcing the model to maintain consistent predictions before and after disturbances, robustness and generalization ability can be enhanced to achieve better performance. Moreover, prototype-based consistency regularization [31] is based on prototype learning to mine unlabeled data category information and uses a consistency regularization constraint model for semi-supervised semantic segmentation prediction. Due to the complex terrain relationships in remote sensing images, they are susceptible to various factors that may generate noise, leading to unstable model training. Focusing on remote sensing images, Lu et al. [33] proposed an adaptive reweighting strategy (WSCL) based on an entropy graph, gradually filtering out noise in predictions, avoiding model interference from noise, and ensuring the stability and positivity of the training process. The UniMatch V2 [36], using channel level complementary masks, splits a feature stream into two sets of non-overlapping, semantically complementary sub features, thereby constructing texture and structure dual perspectives within the same image. This provides a more informative and low redundancy augmented space for weak strong consistency regularization. However, there is still a barrier between labeled and unlabeled data. The complex terrain features of remote sensing images and the lack of labeled samples can easily lead to incorrect predictions of unlabeled data by the model, resulting in failed consistency regularization and negative impacts on the model.

For remote sensing images, in order to integrate and enhance the advantages of self-training and consistency regularization methods to improve segmentation accuracy, two problems must be solved.

1) How to prevent consistency regularization failure during model training due to incorrect predictions of unlabeled data. Due to the complexity of remote sensing images, labeled datasets only contain certain subclasses of one type of land cover, resulting in the remaining subclasses in unlabeled datasets being unable to be covered by labels. The barrier between labeled and unlabeled datasets can lead to incorrect predictions during the training process, resulting in failed consistency regularization and unsupervised information misleading the model's learning process.
2) How to improve the credibility of generating pseudo-labels. When the model predicts unlabeled data, due to the scarcity of labeled data, the information learned by a single model is not sufficient, and the vision is not comprehensive. If the model extracts information from pseudo-labels without filtering, the confidence of pseudo-labels is greatly reduced, and errors continue to accumulate in the model, leading to incorrect training processes and further expanding the model's errors.

Based on this, we design a semi-supervised semantic segmentation method applicable to remote sensing images, with the main contributions presented below.

1) During the model training process, the enhanced guided learning strategy that utilizes labeled data to guide the prediction direction of unlabeled samples effectively alleviates the problem of consistency regularization failure caused by incorrect predictions. In addition, data augmentation in enhanced guided learning enriches the features and relationships of land features and improves the coverage of label categories. Unsupervised loss dynamic screening is used to remove the noise generated during data augmentation, significantly improving the model's generalization ability and anti-interference robustness.
2) When performing pseudo-label prediction and generation on unlabeled data, the Bayesian formula is first used to optimize the probability matrix and remove low-confidence regions, considering the different prediction accuracies of the dual model for different categories. Subsequently, the similarity matrix is obtained through a dual model architecture, which fully integrates the advantageous characteristics of different models to accurately locate and remove uncertain areas in pseudo-labels, generating high-quality and high-confidence pseudo-label results.

The remaining sections of this article are organized as described subsequently. In Section II, we introduce relevant research. Section III elaborates on the work put forward in this article. In Section IV, we present the experimental dataset, the experimental setup, and the results and analysis of the experiments. Finally, in Section V, we offer a summary.

II. Related Work

A. Self-Training

As an efficient method to produce pixel-level labels for unlabeled data, self-training has been broadly implemented in semi-supervised semantic segmentation tasks. Early frameworks [25] used models trained on labeled data to predict unlabeled samples, generating pseudo-labels via entropy regularization, but low confidence in these labels hindered unlabeled data utilization. Later work [37] adopted multitask models to extract pseudo-mask statistical features, optimizing segmentation networks through consistency constraints and

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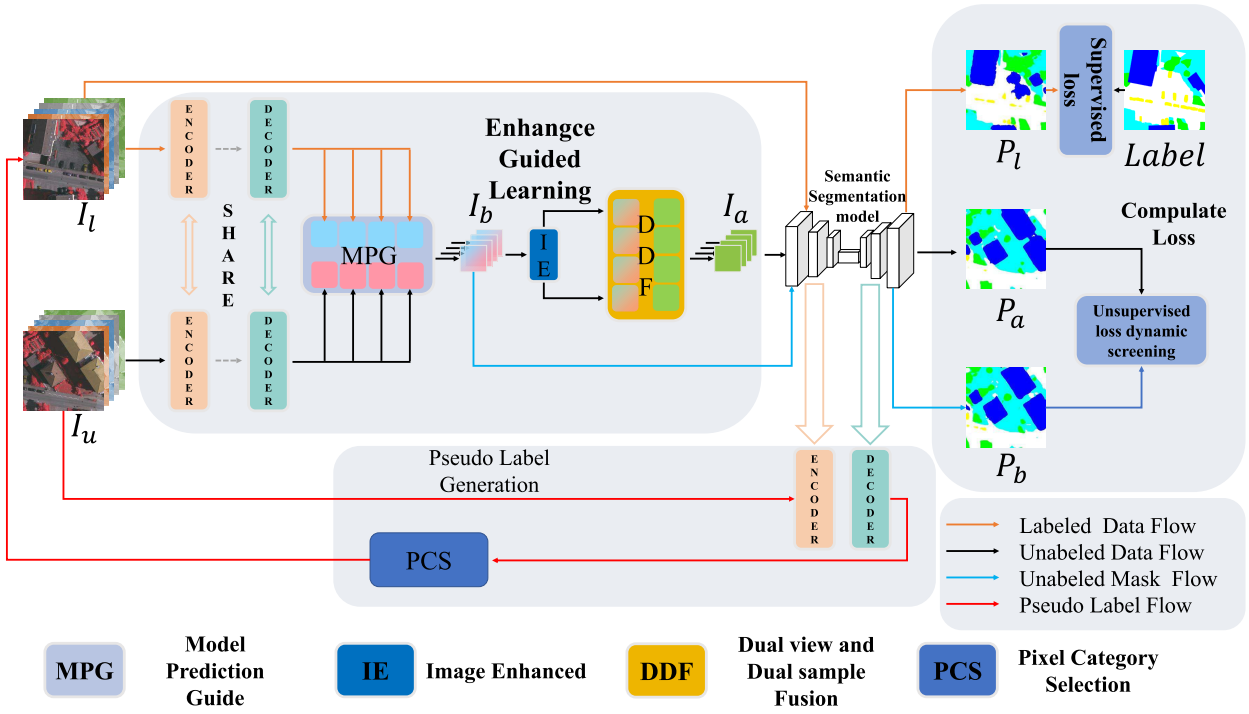


Fig. 1. Overview of our method (batch size is four) is as follows: The MPG module is used to guide the combination of unlabeled and labeled data; then, DDF is applied to enhance the mixed image; and dynamic screening of unsupervised loss is adopted to address the issue of unsupervised loss.

iterative pseudo-mask refinement. To tackle low pseudo-label confidence, ST++ [26] and LSST [27] introduced phased optimization and filtering strategies, respectively. Recent research [38] has improved the quality of pseudo-labels by labeling dynamic pseudo-label sampling, distribution matching, and adaptive threshold updating in datasets. However, due to the limitations of single-model cognition in remote sensing images, it is difficult to detect the erroneous predictions generated by themselves. Considering the performance differences between models, the PCS module uses the Bayesian formula to calculate the prediction probability matrix, allowing probabilities to be compared under the same premise. Dual mode prediction is used to locate and filter uncertain pixels, greatly improving the confidence of pseudo-labels.

B. Consistency Regularization

Consistency regularization is first introduced in semi-supervised learning by Sajjadi et al. [28], who proposed an unsupervised loss function and used data augmentation, dropout, and random pooling to minimize prediction discrepancies of the same sample across multiple network passes for unlabeled data learning. Building on the concept of CutMix [39], ClassMix [29] utilizes a mean teacher architecture with exponential moving average (EMA) to modify the teacher network's parameters. As a result, it can generate consistent prediction labels for images, effectively enhancing the exploitation of unlabeled data. WSCL [33] data augmentation enriches unlabeled data and filters noise and interference through entropy. However, when faced with unlabeled data, the model tends to make incorrect predictions, leading to failed consistency regularization and misleading unsupervised information. In addition, the noise generated during data augmentation can also affect the performance of the model. The MPG module we designed guides the model to make predictions in the correct direction through labels, alleviating the misleading caused by incorrect predictions. The unsupervised loss dynamic filtering module accurately locates and removes noise to reduce its impact.

III. Method

Enhanced guided learning, unsupervised loss dynamic screening, and pseudo-label generation by pixel category selection (PCS) are all parts of the semi-supervised semantic segmentation method we designed. The overall steps are shown in Fig. 1.

A. Enhanced Guided Learning

In order to break down the barrier between labeled and unlabeled data and prevent consistency regularization from failing and leading to unsupervised information misleading the model, we have designed an enhanced guided learning strategy to reduce the occurrence of erroneous predictions and enriched the unsupervised information through dual view and dual sample fusion (DDF). The detailed program is shown in Figs. 2 and 3, as follows.

1) Dataset Definition and Preparation: In the task of semi-supervised semantic segmentation of remote sensing images, we have a labeled dataset I_l containing a small amount of labeled data $x^l \in R^{W\times H\times 3}$, corresponding labels $y^l \in R^{W\times H}$, and an unlabeled dataset I_u containing numerous unlabeled data $x^u \in R^{W\times H\times 3}$, where W and H, respectively represent the size of the data.

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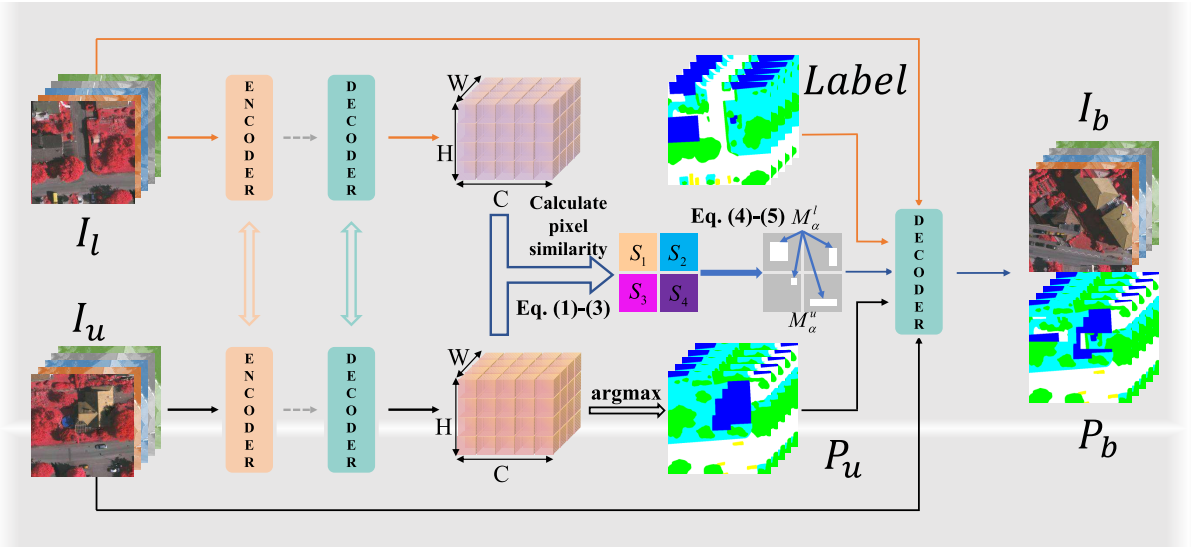


Fig. 2. MPG: randomly select labeled data and unlabeled data, predict them based on the training model, and calculate the similarity matrix S. Divide the graph into blocks, calculate the exchange area ratio separately, and obtain guidance data by mixing the data according to the ratio.

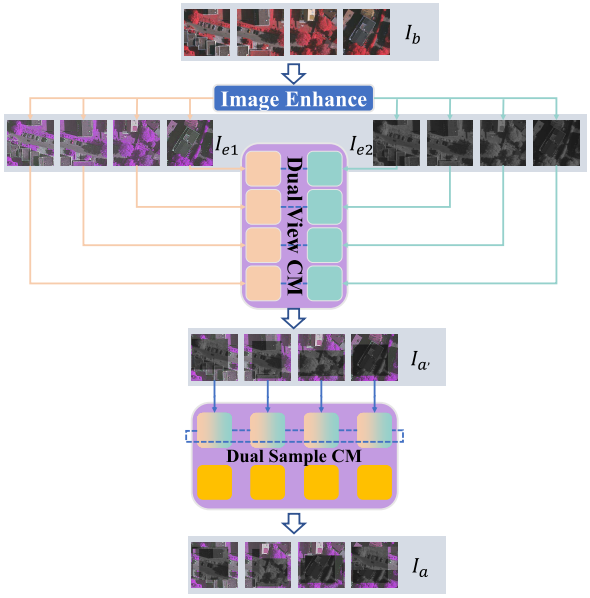


Fig. 3. DDF: I_{e1} and I_{e2} represent different enhanced samples, and CM represents randomly executing CutMix. Obtain $I_{a'}$ and I_a through DDF, respectively.

Let us combine the labeled and unlabeled dataset into $\{x^u, x^l, y^l\}$ before each training cycle and perform random one-on-one guidance to enhance data randomness.

2) Model Prediction and Similarity Calculation: The model's predictions can be guided correctly by the labels. Pasting labeled images onto unlabeled ones serves as a positive random perturbation. For inputs x^l and x^u, the guiding force should be determined based on their similarity. If the terrain features of x^l and x^u differ greatly, the guiding force can lead to misunderstandings; if they are similar, it aids learning. Thus, evaluating the similarity between x^l and x^u is essential.

Therefore, we must evaluate the similarity between x^l and x^u. The specific similarity score matrix $S_{lp} \in \mathbb{R}^{W\times H}$ is solved using the following formula:

$$S_{lp} = \frac{P_l \otimes P_u}{\|P_l\|_\Gamma \odot \|P_l\|_\Gamma} \tag{1}$$

where $P_l, P_u \in \mathbb{R}^{W\times H\times C}$ are the model's prediction probability matrices for x^l and x^u with C categories. The $\otimes$ is a pixel-wise C-dim vector dot product, the $\odot$ is the element-wise matrix multiplication, and the $\|\|_\Gamma$ is the pixel-wise vector length. In (1), the dot product operation quantifies the directional consistency between two vectors and normalizes them to the interval of [0,1], effectively evaluating the similarity between two probability matrices pixel by pixel.

Since individual pixels are easily affected, we use the following formula to expand the field of view and improve similarity stability:

$$S_{lp}(w,h) = \frac{1}{r^2}\sum_{i=1}^{r}\sum_{j=1}^{r} S(w+i-r, h+j-r) \tag{2}$$

where r is the local field of view.

Finally, normalization is used to eliminate the influence of maximum and minimum extreme values, and the specific formula is as follows:

$$S_{lp} = \frac{S_{lp} - \min(S_{lp})}{\max(S_{lp}) - \min(S_{lp})}. \tag{3}$$

3) Guidance: Faced with the wide coverage of remote sensing images, significant differences in land cover types, and instability issues in early model training, we divide the image and similarity score matrix S_{lp} into four blocks $S_{lp}^i, i \in [1,4]$ and $\overline{S_{lp}^i}$ is the average value of S_{lp}^i.

For each block, we use the following formula to calculate the similarity α_i for matrix S_{lp}^i separately:

$$\alpha_i = \left(\left(1 - \overline{S_{lp}^i}\right)\cdot \text{rand}(0,1) + \overline{S_{lp}^i}\right)^2 \tag{4}$$

where rand(0,1) refers to generating a random number between 0 and 1. α_i combines the local similarity and random perturbation of labeled and unlabeled data with the dynamic balance

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of terrain guidance, thereby enhancing adaptability and anti-interference capability.

Then a binary matrix $M^l_{\alpha_i}$ of size $W \times H$ is generated, with 1-valued areas accounting for α_i of the total. Subsequently, $M^u_{\alpha_i} = 1 - M^l_{\alpha_i}$ is derived, and the two matrices are combined based on their original positional relationship.

We obtain the guidance dataset I_b, which includes guidance data x^b and prediction probability matrix P_b, through the following formula:

$$\begin{aligned} x^b &= M^l_\alpha \odot x^l + M^u_\alpha \odot x^u \\ P_b &= M^l_\alpha \odot P_l + M^u_\alpha \odot P_u. \end{aligned} \tag{5}$$

Due to the presence of labels, P_l has higher accuracy and correctness, guiding the model to make predictions in the correct direction, thereby alleviating the failure of consistency regularization and making it easier for the model to learn unsupervised information in x^b.

4) DDF: In order to provide the model with as rich land cover relationships and features as possible, we designed dual view and DDF for data augmentation using CutMix [29], as shown in Fig. 3.

In each epoch, we perform two independent data augmentation operations on each image in I_b, including color jitter, grayscale processing, blur processing, and edge enhancement, to obtain two different augmentation datasets I_{e1} and I_{e2}, respectively. And mix it to obtain I_a with rich data by using the following formula:

$$\begin{aligned} I^k_{a'} &= \mathrm{CM} \odot I^k_{e1} + (1 - \mathrm{CM}) \odot I^k_{e2} \\ I^k_a &= \mathrm{CM} \odot I^k_{a'} + (1 - \mathrm{CM}) \odot I^{k+1}_{a'} \end{aligned} \tag{6}$$

where CM is a binary matrix that can mix data with an equal area, and k represents the kth data in the dataset.

The DDF method can enhance data by mixing data and introducing regional perturbations and random noise, alleviate the problem of low label coverage through mixed data, and provide rich land cover relationships and features. Collaborative guided learning can guide the model to make predictions in the correct direction, alleviate the problem of consistency regularization failure, and enable the model to better learn and understand the features of remote sensing images.

B. Unsupervised Loss Dynamic Screening

Although DDF enhances the training set, it also injects additional noise and perturbations. Learning all unsupervised information without screening can mislead the model and affect segmentation accuracy. To this end, we propose an unsupervised loss funnel that can dynamically screen gradients, preserve reliable information, and suppress difficult to learn noise.

1) Similarity Calculation of Changes: Difficult to learn noise can lead to significant differences in the network's prediction of images before and after enhancement. Using (1)–(3), we solve the similarity matrix S_{ab} of the prediction probability matrixes P_b (before enhancement) and P_a (after enhancement), thereby locating noisy pixels.

2) Dynamic Threshold Calculation and Unsupervised Loss Solving: Excessive noise hinders loss function convergence, while insufficient noise limits the model's anti-interference capability. To balance, we retain more pixels initially to speed up learning. As training progresses and prediction improves, we gradually reduce input noise and interference, enhancing anti-interference ability and preventing random disturbances from affecting performance.

The precise formula for determining threshold adjustments is as follows:

$$\begin{cases} \gamma_i = \mathrm{sort}\left(S^i_{ab}.\mathrm{flatten}\right)[W \cdot H \cdot (1 - \tau_i)] \\ \mathrm{s.t.}\tau_i = \tau_n \left(\dfrac{\tau_0}{\tau_n}\right)^{\frac{i}{n}} \end{cases} \tag{7}$$

where i represents the current training round, n is the total round. τ_0 and τ_n are the initial and final data selection ratios, respectively, while τ_i is the pixel ratio of the ith epoch. As the training progresses, τ_i controls the proportion of pixels introduced into unsupervised computing, and its decrease gradually reduces the introduction of noise and interference, as shwon in Fig. 4.

Ultimately, our unsupervised loss can be expressed as follows:

$$L_u = \begin{cases} \dfrac{1}{N_u} \displaystyle\sum_{i=1}^{N_u} L_{\mathrm{ce}}\left(\arg\max\left(P_a\right), P_b\right), & \text{if } S^i_{ab}(w,h) > \gamma_i \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where N_u represents the quantity of unlabeled data and L_{ce} is expressed as follows:

$$L_{\mathrm{ce}}(y,x) = -\frac{1}{H \times W} \sum_1^H \sum_1^W \sum_1^C y_{h,w} \cdot \log x_{h,w,c} \tag{9}$$

which is cross entropy loss.

This module identifies and removes interference based on the model's training status, ensuring consistent predictions before and after data augmentation. This boosts the model's generalization and stability, and enhances the reliability of unsupervised information.

C. Pixel Category Selection

Enhanced guided learning and unsupervised loss dynamic screening are model training strategies. We train pseudo-label generation models with these strategies and then train the final model using the generated pseudo-labels. Specific training steps are detailed in Section III-D.

Models inevitably produce incorrect predictions when predicting unlabeled data, and the accumulation of errors on the model may further mislead the model. Therefore, we plan to filter out pseudo-labels to improve their credibility and quality. Therefore, we designed a PCS strategy and proposed a method for generating pseudo-labels using dual models, as shown in Fig. 5.

1) A Posteriori Prediction Probability and Preliminary Filtering: Convolutional models recognize small object details, while transformer architectures capture broad contextual relationships. To combine these strengths, we selected Farseg++

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[40] as Model 1 and CMTFNet [41] as Model 2. Other models can be used based on specific needs.

For each unlabeled image, the predicted probability distribution matrices $P_1, P_2 \in \mathbb{R}^{W\times H\times C}$ are obtained through the preliminary training of Models 1 and 2.

However, the performance differences between models can lead to different confidence levels in predicting probabilities, so we cannot directly compare probabilities. To this end, we consider using the IoU of the test set model as the likelihood probability, applying Bayesian formula to re-evaluate the prior probability matrix and improving the overall reliability of the probability matrix by filtering and eliminating low confidence probabilities. The following formula takes Model 1 as an example:

$$P_1(h,w,c) = \begin{cases} P_1(h,w,c)\cdot \mathrm{IoU}_c^1, & P_1(h,w,c) > \delta^1 \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

where the threshold calculated by $\delta^1 = \mathrm{sort}(P_1.\mathrm{flatten})[H\times W\times C\times(1-(\mathrm{IoU}^1/100))]$ and IoU_c^1 is the result of evaluating Model 1 for the cth class, considering the confidence differences of different class models. $\overline{\mathrm{IoU}^1}$ is the average value of IoU_c^1, ensuring that labels are not imbalanced due to class imbalance. δ^1 is the threshold value for Model 1 to select high-confidence regions. After filtering, the posterior probability matrices P_1 and P_2 with high confidence are obtained.

2) Dual Model Fusion Generates Pseudo-Labels: After the first round of probability calibration, the probability matrices P_1 and P_2 can be compared at the same level. We compare P_1 and P_2 and select the category with the highest probability as the label at (w,h). We can obtain the pseudo-label Mask $\in \mathbb{R}^{W\times H}$ through the following formulas:

$$\mathrm{Mask}(w,h) = \begin{cases} M_1(w,h), \text{if } P_1(h,w,M_1(w,h)) > P_2(h,w,M_2(w,h) \\ M_2(w,h), \text{otherwise} \\ s.t. M_1 = \arg\max(P_1), M_2 = \arg\max(P_2) \end{cases} \tag{11}$$

where $M_1, M_2 \in \mathbb{R}^{W\times H}$ obtain the category index with the highest probability value at positions (h,w) of P_1 and P_2 through the argmax function.

3) Uncertain Pixel Removal: Without labels, it becomes difficult to determine the specific category of pixels when there is a significant difference in the predicted distribution between the two models. However, similarity analysis can be used to identify and eliminate these uncertain pixels. So, we calculate the score matrix S_{12} based on (1)–(3) for P_1 and P_2, and then use the following formula to obtain a binary matrix MS of size $W \times H$:

$$\mathrm{MS}(w,h) = \begin{cases} 1, S_{12}(w,h) > \varepsilon \\ 0, \text{otherwise} \end{cases} \tag{12}$$

where ε is the threshold for filtering pseudo-labels. Then, we use the following formula: Mask = Mask $\odot$ MS to obtain the final pseudo-labels.

Our PCS module integrates the advantages of convolutional and transformer models, effectively reducing the accumulation of errors in pseudo-labels through filtering and removing difficult to determine pixels and noise through pixel-level differences between P_1 and P_2, thereby improving the confidence and quality of pseudo-labels.

D. Training Process

The overall training process we designed is shown in Fig. 1, and the specific steps are as follows.

1) Definition of Loss Function: Our loss function is as follows:

$$\begin{cases} L = L_l + L_u \\ \text{s.t. } L_l = L_{\mathrm{ce}}(y^l, P_l) \end{cases} \tag{13}$$

where L_u is calculated through (8).

2) Pseudo-Label Generation Models Training: By independently training the pseudo-label generation networks FarSge++ and CMTFNet through strategies enhanced guided learning and unsupervised loss dynamic screening, Models 1 and 2 are obtained.

3) Pseudo-Label Generation: We generate high-confidence pseudo-labels using Models 1 and 2 with preliminary segmentation capabilities through the PCS module. Then, we mix the pseudo-labels, labeled dataset I_l, and unlabeled dataset I_u to obtain the entire dataset I.

4) Final Model Training: We then replace both the labeled and the unlabeled datasets with our overall dataset I and train the final model through strategies enhanced guided learning and unsupervised loss dynamic screening independently.

IV. Experimental Results

A. Datasets Description and Experimental Setup

1) Datasets Description: The MER dataset consists of high-resolution images of the Martian surface, with unclear colors and blurred edges. The DFC22 and GID-15 datasets have a high spatial resolution, a wide range of land cover types, and complex relationships. The iSAID and Vaihingen datasets have high resolution and focus on urban images. As shown in Table I, we use diverse datasets, including near-grayscale and color images with spatial resolutions from 0.3 to 400 cm/px [46]. And the scene is rich, including natural landscapes, urban landscapes, and Mars remote sensing images.

The spatial resolution significantly affects the information granularity and category separability of remote sensing images. Low spatial resolution leads to detail masking, spectral overlap, and smoothing of interclass differences, blurred edges, and the model requires a large receptive field for overall inference, which is prone to under segmentation. Although high spatial resolution significantly improves pixel-level detail granularity, it amplifies intraclass texture and material differences, resulting in a significant increase in intraclass noise and a decrease in intraclass separability, requiring the model to have stronger texture discrimination ability.

2) Experimental Setup: In the experiment, we uniformly cut the images into sizes of 512 $\times$ 512. The experiment in this article is run on a Windows 10 system, using an RTX3090 24 GB GPU and 32 GB RAM. When evaluating classification performance, we use IoU as the evaluation metric and the

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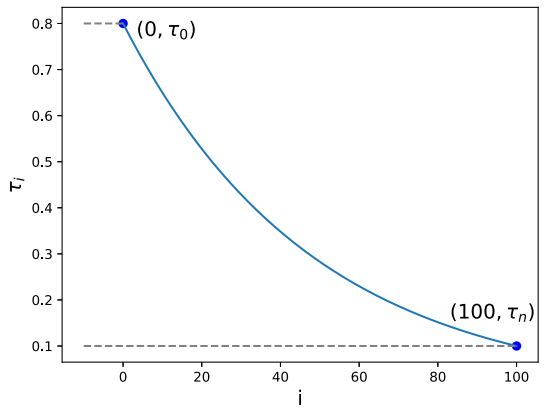


Fig. 4. Variation of τ_i (pixel selection ratio for unsupervised loss calculation) with training epochs in the unsupervised loss dynamic screening module. The horizontal axis is training epochs (0–100, total $n = 100$), and the vertical axis is τ_i value. The curve decreases from initial $\tau_0 = 0.8$ to final $\tau_n = 0.1$, balancing early unsupervised information absorption and late noise suppression for remote sensing semi-supervised semantic segmentation.

TABLE I
SPECIFIC INFORMATION OF DATASET

Dataset	Classes	Size	Spatial Resolution(cm/px)	Satellites
MER[42]	9	1064	0.3–1.4	Navcam Pancam
DFC22[43]	12	766	50–100	-
iSAID[44]	15	1411	80	Gaofen-1 Gaofen-2
GID-15[45]	15	3420	400	Gaofen-2
Vaihingen[1]	5	1194	9	Leica ALS50

TABLE II
EXPERIMENTAL RESULTS OF EACH MODULE ON MER DATASET

Baseline	MPG	DDF	Dynamic screening	MIoU
✓				44.12
✓		✓	✓	53.03
✓	✓		✓	50.54
✓	✓	✓		50.77
✓	✓	✓	✓	53.79

CMTFNet and FarSeg++ models are combined with ResNet-50 as segmentation models.

The hyperparameters are kept constant, with a batch size of 16. Stochastic gradient descent (SGD) is employed for model iteration. The initial learning rate is configured as 0.001. As the training proceeds, the learning rate decays following the formula $l_r = 0.001 \times (1 - (\text{iter})/\max_\text{iter})^{0.9}$. The number of training epochs is set to 100, and the size of the cropped pixels in the training image is set to 321×321.

In comparative experiments, we compared with other excellent semi-supervised semantic segmentation strategies for remote sensing images, including the following strategies: AdvSS [47], s4GAN [17], ClassMix [29], ST++ [26], LSST [27], WSCL [33], UniMatch V2 [36], and "SupOnly" uses Deeplabv2 with resnet101 for training only under supervised loss conditions. Part of the experimental data comes from the previous studies [27], [33], and the specific visualization details are shown in Fig. 6.

B. Comparison and Analysis of Hyperparameters

We tested two key hyperparameters using the control variable method due to the modules' independence. Here are the experiment details.

1) τ: τ appears in (7) as the filtering ratio hyperparameter in the unsupervised loss dynamic screening module, which determines the proportion of pixels involved in unsupervised loss calculation in each round of training. The magnitude of variable τ affects the extent to which the model is affected by noise. A too large τ affects the convergence of the loss, while a too small τ weakens the model's anti-interference ability.

In order to fine-tune the model in the later stage, we set τ_n to a smaller value of 0.10. The τ appearing in the experiment refers to τ_0. For τ_0, we adjusted it from 0.50 to 1.00 in increments of 0.10, as shown in Fig. 7. The graph shows that, for each dataset, the performance is optimal when τ_0 is 0.80. When τ_0 is 0.50, most of the unsupervised information is filtered out, resulting in limited information obtained by the model. With the increase of τ_0, the model's performance experiences a rapid enhancement. This is attributed to the substantial growth in the information acquired by the model. When τ_0 is too large, the model is inundated with a great deal of noise and interference information, which results in a decline in the model's performance. So, the model performs best when τ_0 is 0.80.

2) ε: ε in (12) determines the pseudo-label generation module's score threshold for pseudo-label filtering. Pixels with scores less than ε are filtered out. The magnitude of variable ε affects the final confidence level of this pseudo-label. The higher the ε, the less information the pseudo-label retains and the higher the credibility. The lower the ε, the more information the pseudo-label retains and the lower its credibility.

As shown in Fig. 8, we can see from the experiment that, when ε increases from 0.60 to 0.85, the confidence of the pseudo-labels increases and the performance of the model improves; when ε continues to rise from 0.85, the model performance tends to stabilize but slightly decreases. So we believe that, when ε is 0.85, the performance of the model is best.

C. Ablation Experiments

To validate the effectiveness of our designed modules, model prediction and similarity calculation, and guidance (MPG) in Section III-A, dual view and DDF in Section III-A, and unsupervised loss dynamic screening (dynamic screening) in Section III-B, we compared the roles of each module based on their performance on the MER dataset. To guarantee the fairness of the experiment, parameters other than those in the comparison module remain unchanged. The detailed experimental results are presented in Table II.

1) MPG Module: Compared with other modules, the MPG module uses labeled data to guide the model to predict unlabeled data in the correct direction, thereby improving the effectiveness of consistency regularization. From the results, it can be seen that the experimental results without the MPG module decreased by 0.76% in terms of the results obtained

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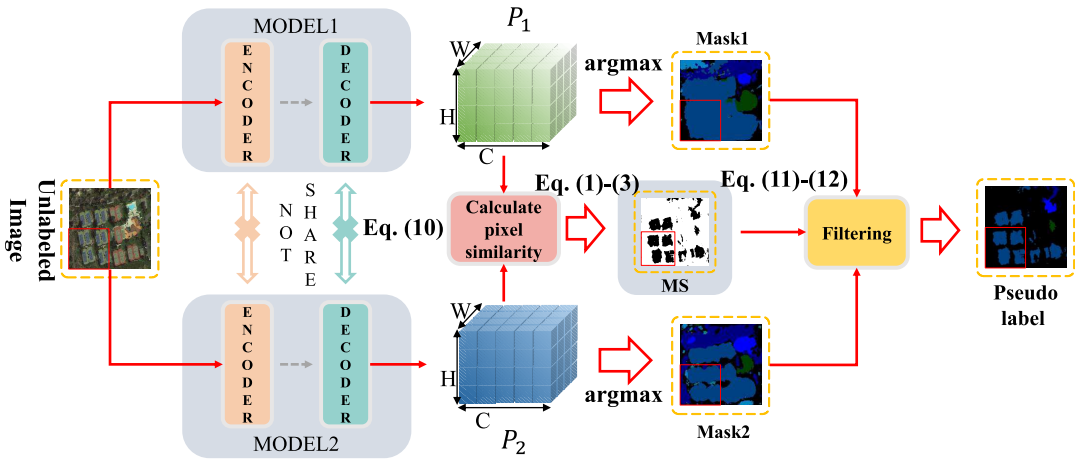


Fig. 5. Overview of the pseudo-label generation process. We use two independent pre-trained models to predict unlabeled data, obtain two prediction probability matrices, filter the matrices, calculate similarity to remove low-confidence areas, and compare to obtain the final pseudo-labels.

TABLE III
VARIOUS IOUS OF MER DATASET

Labeled 1/8(106)	SupOnly	AdvSS	s4GAN	ClassMix	ST++	LSST	WSCL	UniMatch V2	Ours
Martian Soil	*4.54*	1.86	4.06	2.84	2.04	3.87	3.45	**5.44**	2.46±0.55
Sands	45.47	48.26	46.31	47.35	51.63	52.30	**54.84**	*52.82*	45.07±1.22
Gravel	63.68	67.26	63.80	67.64	66.04	68.27	69.91	**73.39**	*72.22±1.67*
Bedrock	50.84	57.96	50.72	52.57	56.98	60.33	*62.81*	61.44	**68.93±1.01**
Rocks	29.40	29.62	21.83	26.56	26.22	29.59	*35.97*	33.83	**37.05±0.63**
Tracks	45.09	47.47	44.65	39.30	50.27	53.98	*56.91*	52.89	**59.34±0.33**
Shadows	25.60	25.49	24.50	27.65	25.40	25.88	26.13	*29.40*	**33.92±0.88**
Unknown	63.24	62.52	63.83	65.41	65.56	65.28	*75.97*	**78.51**	72.26±1.71
Background	64.80	67.57	75.43	74.28	83.85	87.58	80.93	*83.22*	**91.82±2.00**
MIoU	43.63±0.14	45.34±0.52	43.90±0.75	44.84±0.72	47.55±0.80	49.68±0.33	51.88±0.17	*52.33±0.55*	**53.67±1.14**

using this module. The main reason is that the lack of the MPG module reduces the model's understanding of unlabeled data, consistency regularization experiences a certain degree of failure, and there is a lack of unsupervised information acquisition.

2) DDF Module: The dual view DDF module is used to mix and enhance data, adding a certain amount of learnable noise and interference. Reasonable noise interference can improve the stability and noise resistance of the model. The results indicate that, when the module is not used, the performance of the model significantly decreases by 3.25%. This is mainly due to the imbalance of samples in the data and the lack of additional terrain information.

3) Unsupervised Loss Dynamic Screening Module: The primary function of this module is to filter and remove random disturbances and interference that cannot be learned to prevent low-quality information from interfering with the model. The data presented Table II shows that, without this module, there is a substantial 3.02% decline in the model's prediction accuracy. The fundamental reason is the lack of screening for unsupervised information. This leads to the model learning too much random noise, making it difficult for the loss to converge, thereby affecting the model's predictions.

4) PCS Module: We evaluated the improvement in pseudo-label accuracy of the PCS module through comparative experiments on multiple datasets. The results in Fig. 9 indicate that the pseudo-labels generated by the PCS module improve the accuracy and confidence of the pseudo-labels. This makes pseudo-labels more accurate and reliable, reducing the risk of error accumulation and model misguidance.

D. Experimental Result on MER Dataset

The MER dataset's low resolution is only 0.3–1.4 cm/px, and the rich details and near-grayscale images pose segmentation challenges. We compare our method with mainstream semi-supervised remote sensing image semantic segmentation methods to verify its effectiveness, with quantitative results in Table III.

AdvSS and s4GAN use adversarial methods but suffer from unstable GAN sample quality and low accuracy. ClassMix generates new samples by mixing but performs poorly due to limited samples. ST++ and LSST enhance pseudo-label confidence yet lack generalization from insufficient semi-supervised information. WSCL employs consistency regularization, leveraging weakly and strongly enhanced data views to learn from unlabeled datasets, and attains the highest accuracy in *Sands* categories. UniMatch V2, with a more concise single stream complementary enhancement framework, has achieved significant improvements in segmentation accuracy in scenarios with a wide range of categories. However, it has not yet been specifically optimized for the characteristics of remote sensing

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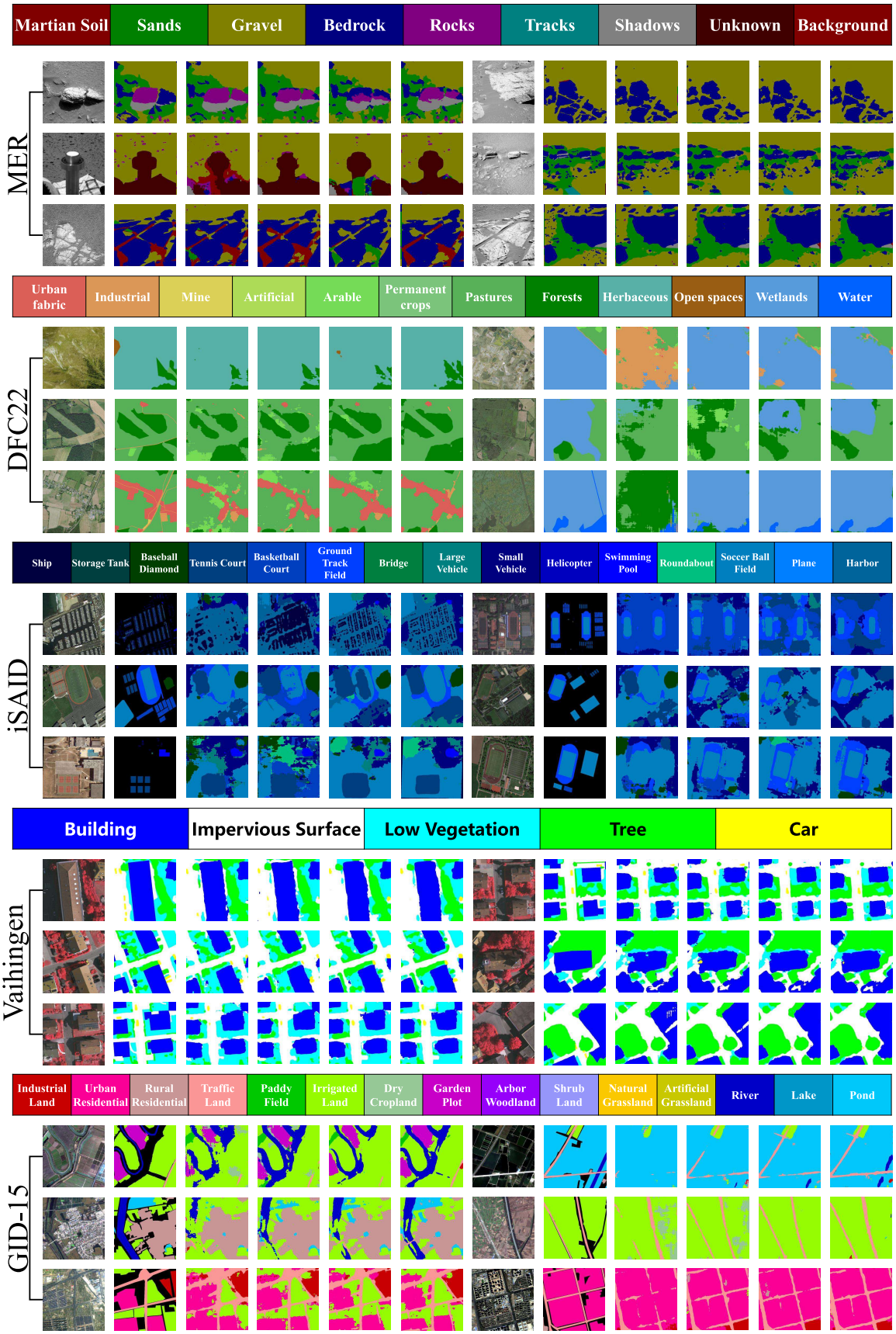


Fig. 6. Visual comparison results of this method with other advanced semi-supervised image segmentation methods with high segmentation accuracy. (Left to right) Input, ground truth, LSST, WSCL, UniMatch V2, and ours.

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TABLE IV
VARIOUS IOUS OF DFC22 DATASET

Labeled 1/8(76)	SupOnly	AdvSS	s4GAN	ClassMix	ST++	LSST	WSCL	UniMatch V2	Ours
Urban	54.10	58.63	58.67	56.77	60.18	60.54	61.19	*61.27*	**63.87±2.02**
Industrial	29.54	29.55	28.12	29.68	33.58	38.44	*40.66*	35.68	**41.79±0.85**
Mine	0.81	0.08	0.09	*3.32*	-	5.11	5.50	**10.89**	8.13±5.66
Artificial	7.27	7.21	10.17	1.06	15.39	*17.15*	16.16	14.02	**17.35±1.25**
Arable	10.03	7.44	12.56	2.69	11.99	9.76	10.35	**15.17**	*10.75±0.97*
Permanent	43.37	49.33	55.37	36.92	**59.95**	52.29	*58.13*	50.63	55.95±1.07
Pastures	46.84	52.42	44.85	53.91	54.25	52.81	53.14	*55.73*	**59.35±0.82**
Forests	61.19	59.68	64.76	65.35	63.72	63.51	**70.21**	65.59	*69.86±0.67*
Herbaceous	32.88	36.35	37.01	36.17	35.58	36.11	*40.44*	**40.75**	39.44±1.66
Open	22.07	*24.48*	18.42	0.38	1.86	5.49	38.26	**40.03**	7.03±3.32
Wetlands	8.20	5.27	0.09	3.16	0.83	6.55	*31.58*	20.66	**60.34±0.88**
Water	7.35	4.87	16.96	4.59	15.62	23.53	*30.34*	24.35	**31.12±0.67**
MIoU	26.97±0.86	27.94±0.80	28.92±0.83	24.50±1.69	29.41±0.98	30.94±0.61	*38.00±0.76*	36.23±0.82	**38.75±0.72**

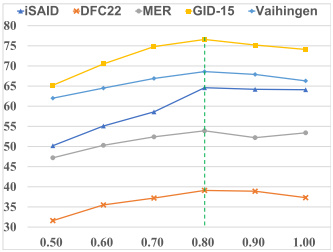


Fig. 7. Results with different hyperparameter values τ on DFC22, iSAID, MER, GID-15, and Vaihingen datasets are compared using the MIoU as the score.

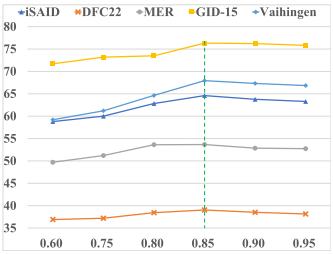


Fig. 8. Results with different hyperparameter values ε on DFC22, iSAID, MER, GID-15, and Vaihingen datasets are compared using the MIoU as the score.

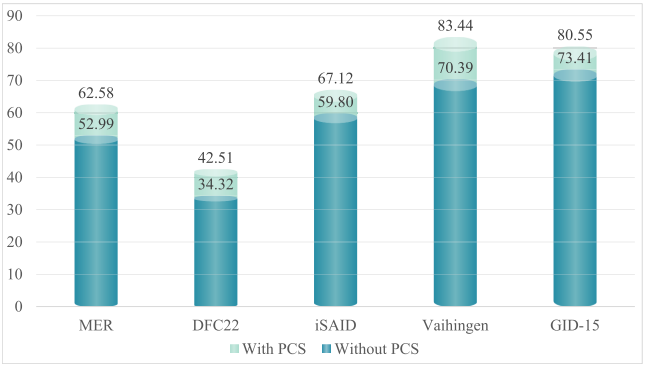


Fig. 9. Use of MIoU as a score to compare the accuracy and confidence of pseudo-labels with and without PCS modules.

images and still has considerable potential for improvement in the future.

The segmentation difficulty of the MER Mars remote sensing dataset is relatively high, with highly overlapping features of the *Martian soil*, *Sands*, and *Gravel*. The surface needs to indicate the need to distinguish finer texture features for differentiation. Our method achieved the highest or second highest accuracy in various categories of the Mars remote sensing dataset, reaching 53.67% MIoU, proving the superiority of our method in complex remote sensing datasets. It's ability to analyze similar category features has also provided effective assistance for semi-supervised semantic segmentation in space exploration and planetary science [48].

E. Experimental Result on DFC22, iSAID, Vaihingen, and GID-15 Datasets

We analyze the segmentation results for other datasets. The detailed displayed results in visual form are presented in Fig. 6.

The DFC22 dataset focuses on remote sensing images of natural scenes, with a large range of spatial resolution variations and difficult segmentation due to unclear boundaries. The iSAID dataset focuses on remote sensing images of urban scenes with moderate spatial resolution, but small land features greatly increase the difficulty of segmentation. Our method has demonstrated high accuracy on the DFC22 and iSAID datasets, as shown in Tables IV and V. However, due to the similarity of classes, the accuracy of similar classes may vary greatly. The categories of *Arable* and *Forests* in the DFC22 dataset, and *Helicopter* and *Plane* in the iSAID dataset are similar and difficult to distinguish. This type of confusion has become a bottleneck that restricts accuracy and is also the core direction for future improvement.

The GID-15 and Vaihingen datasets have significant spatial resolution differences (9 and 400 cm/px, respectively), covering both urban and natural areas with rich scenes. Our method has excellent segmentation ability on these two datasets, as shown in Tables VI and VII. We achieved the highest segmentation accuracy in most categories of GID-15 and all categories of the Vaihingen dataset, achieving MIoU of 63.61% and 76.48%, respectively. In these two datasets, the proportion of misclassified categories is relatively low, indicating that it can effectively segment images of different resolutions and terrains. This is attributed to the coordination between the various modules we designed, which improves the utilization

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TABLE V
VARIOUS IOUS OF ISAID DATASET

Labeled 100	SupOnly	AdvSS	s4GAN	ClassMiX	ST++	LSST	WSCL	UniMatch V2	Ours
Ship	37.28	38.46	40.52	28.59	45.85	**46.83**	39.17	*45.87*	42.16±0.32
ST	4.70	3.95	0.79	1.59	-	1.85	**87.31**	82.33	*86.05±0.77*
BD	62.30	60.24	48.09	48.32	52.44	73.46	59.59	*81.62*	**92.32±0.75**
TC	80.00	73.92	75.52	72.45	83.38	84.69	**90.93**	72.5	*88.16±0.49*
BC	4.95	6.37	9.54	7.76	10.79	15.01	**75.27**	*65.35*	39.31±0.41
GTF	38.85	34.55	34.61	44.00	22.60	42.44	*62.59*	54.01	**64.70±1.65**
Bridge	1.16	0.32	0.38	7.97	-	2.09	14.81	*21.33*	**26.07±4.52**
LV	66.52	66.32	72.13	68.62	73.52	*77.25*	76.36	76.51	**81.19±1.14**
SV	46.76	43.56	44.29	38.07	48.45	54.01	56.85	**66.98**	*63.06±0.69*
HC	-	-	-	-	-	10.60	**22.89**	15.66	*18.17±1.02*
SP	63.24	46.36	47.47	60.13	49.05	70.00	*74.07*	69.43	**75.06±1.40**
RA	65.60	5.52	48.71	52.74	37.88	51.37	35.24	*65.88*	**67.84±1.19**
SBF	48.66	43.16	48.30	43.37	59.90	60.42	62.13	*75.07*	**79.31±1.87**
Plane	38.27	39.01	28.08	28.77	46.18	62.82	**80.52**	70.06	*71.71±0.52*
Harbor	40.31	46.87	45.74	45.05	48.75	51.30	*56.26*	55.65	**60.10±3.57**
MIoU	39.91±1.04	33.91±2.47	36.28±1.66	36.49±1.50	38.59±1.91	46.94±1.17	59.60±1.06	*61.22±0.98*	**63.61±1.21**

TABLE VI
VARIOUS IOUS OF GID-15 DATASET

Labeled 1/8 (342)	SupOnly	AdvSS	s4GAN	ClassMix	ST++	LSST	WSCL	UniMatch V2	Ours
InL	61.27	64.56	62.70	55.22	67.12	67.91	*72.29*	**74.37**	70.44±2.22
UR	72.32	74.88	73.82	69.20	75.96	77.78	*79.08*	77.95	**79.22±2.14**
RR	61.10	64.39	61.98	53.25	62.45	67.23	70.77	**73.76**	*70.95±2.74*
TL	60.50	63.28	62.98	55.40	58.28	64.70	67.10	*71.42*	**72.27±1.15**
PF	65.92	69.18	69.06	43.76	72.31	69.13	80.09	*81.14*	**82.91±1.14**
InL	84.28	84.77	83.85	75.37	84.33	87.24	88.07	*90.51*	**91.38±2.04**
DC	57.92	55.34	52.15	39.09	67.74	69.32	58.28	**90.62**	*78.53±1.39*
GP	15.21	9.84	9.97	2.06	15.58	20.72	*26.40*	22.33	**29.54±4.25**
AW	93.40	92.19	94.77	88.33	94.13	95.88	*97.53*	96.66	**98.00±0.62**
SL	2.59	0.65	0.18	0.01	0.64	2.86	39.20	*47.44*	**57.84±2.69**
NG	84.64	85.79	82.58	23.05	78.91	*90.13*	88.43	84.36	**91.75±0.85**
AG	53.33	53.24	54.48	10.96	45.70	52.97	63.80	**73.58**	*70.93±1.19*
River	82.73	78.75	78.71	67.73	79.41	86.52	*87.66*	75.60	**88.60±0.60**
Lake	74.11	69.68	77.41	67.03	77.50	77.76	82.88	*86.28*	**89.93±2.32**
Pond	58.58	57.37	62.37	33.31	62.48	65.57	69.78	*70.27*	**74.84±1.25**
MIoU	61.86±0.27	61.59±0.48	61.80±0.50	45.58±0.55	62.84±0.36	66.38±0.18	71.42±0.29	*74.42±0.82*	**76.48±1.30**

TABLE VII
VARIOUS IOUS OF VAIHINGEN DATASET

Labeled 1/8(73)	SupOnly	AdvSS	s4GAN	ClassMix	ST++	LSST	WSCL	UniMatch V2	Ours
SF	64.90	67.83	70.77	67.86	72.71	73.73	*75.69*	73.63	**76.64±0.52**
BD	72.13	74.78	77.78	76.22	79.16	79.79	*82.39*	81.45	**82.46±0.60**
LV	44.52	50.18	56.28	48.55	58.97	59.94	61.05	*61.69*	**62.50±0.64**
Tree	63.56	67.31	69.03	67.48	69.22	70.65	70.97	*71.99*	**72.29±0.74**
Car	21.39	25.62	15.37	19.91	19.52	36.32	*43.18*	42.19	**44.27±2.11**
MIoU	53.30±0.23	57.14±0.37	57.85±0.68	56.00±0.39	59.92±0.31	64.09±0.27	*66.65±0.18*	66.19±0.23	**67.63±0.64**

of unsupervised information and the confidence of pseudo-labels.

V. CONCLUSION

We propose a semi-supervised semantic segmentation method for remote sensing images, which uses enhanced guided learning to alleviate the problems of low label coverage and consistency regularization failure caused by incorrect predictions. A new data augmentation method, namely dual view and DDF, has been designed to enrich ground features and unsupervised information and to filter out noise and interfering pixels through unsupervised loss dynamic screening. Meanwhile, in the face of inevitable errors in pseudo-labels, PCS is used to increase the confidence of pseudo-labels and prevent the accumulation of pseudo-label errors in the model. We conducted experiments on different datasets (MER, DFC22, iSAID, GID-15, and Vaihingen). These datasets cover various spatial resolutions, including different planets, cities, and natural scenes. Experiments have shown that this method has strong robustness and adaptability and has great potential in many fields, such as space science exploration and Earth remote sensing monitoring, providing strong support for related research and applications.

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However, the efficiency of this strategy still needs to be improved, and the severe imbalance of similar category samples significantly weakens the segmentation accuracy, which urgently needs to be further addressed. In the future, we plan to introduce large-scale pre-trained models such as SAM and CLIP with rich prior knowledge, and utilize the denoising mechanism of diffusion models to enhance the discriminative and segmentation performance of the models.

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Xifeng Xue is currently pursuing the B.S. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include semi-supervised learning, deep learning, machine learning, remote sensing images, and semantic segmentation.



Hao Zhu (Member, IEEE) received the B.S. degree in physics and photoelectricity engineering and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2013 and 2019, respectively.

He is currently an Associate Professor with the School of Artificial Intelligence and the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include deep learning, multimodal remote sensing image fusion and classification, and remote sensing object detection and tracking.



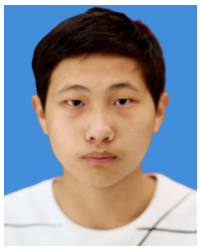
Xiaotong Li received the B.S. degree in electronic information engineering from Harbin Engineering University, Harbin, China, in 2017, and the Ph.D. degree in computer science and technology from Xidian University, Xi'an, China, in 2024.

She is currently doing a post-doctoral research with the Electronic Science and Technology Research Station, Xidian University. Her research interests include deep learning, computer vision, and remote sensing image analysis and understanding.



Jianda Wang is currently pursuing the B.S. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include deep learning, computer vision, remote sensing image processing, and image classification.



Longsheng Qu received the B.S. degree in intelligent science and technology from Xidian University, Xi'an, China, in 2024, where he is currently pursuing the M.S. degree in artificial intelligence with the School of Artificial Intelligence.

His research interests include computer vision, image processing, and deep learning, especially in remote sensing image processing.



Biao Hou (Senior Member, IEEE) was born in Longxian, Shaanxi, China, in 1974. He received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University, where he is currently a Professor. His research interests include dictionary learning, deep learning, and synthetic aperture radar image processing.

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